

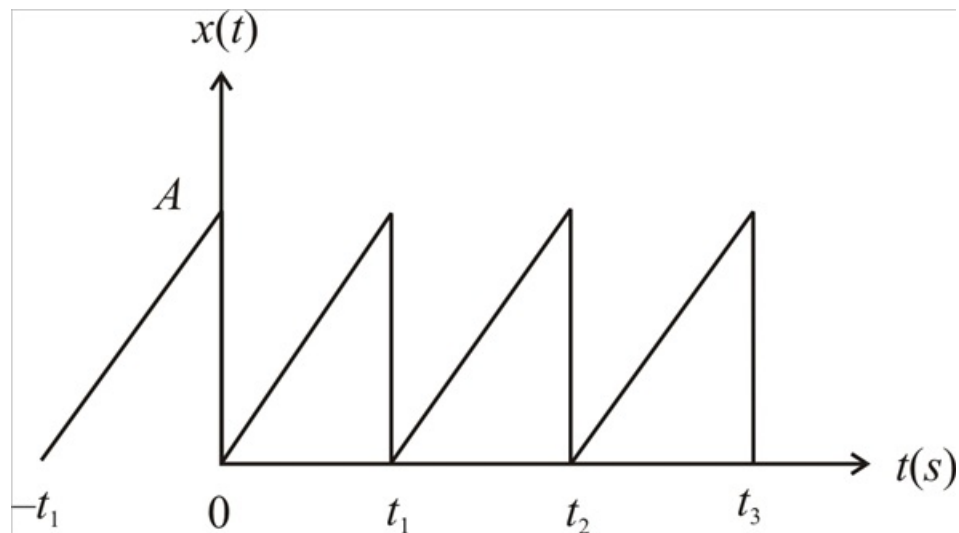
Problem

Design a problem to help other students better understand obtaining the Fourier series from a periodic function.

Step-by-step solution

Step 1 of 6

Let us consider a periodic signal



Step 2 of 6

The function is

$$x(t) = \frac{At}{t_1} \quad 0 < t < t_1$$

Since period T is from 0 to t_1 ,

$$x(t) = \frac{At}{T} \quad 0 < t < t_1$$

$$\omega_0 = \frac{2\pi}{T}$$

Step 3 of 6

The Fourier series coefficients are,

$$a_0 = \frac{1}{T} \int_0^T x(t) dt$$

$$a_0 = \frac{1}{T} \int_0^T \frac{At}{T} dt$$

$$a_0 = \frac{A}{T^2} \left[\frac{t^2}{2} \right]_0^T$$

$$a_0 = \frac{A}{T^2} \left[\frac{T^2 - 0}{2} \right]$$

$$\therefore a_0 = \frac{A}{2}$$

Step 4 of 6

$$a_n = \frac{2}{T} \int_0^T x(t) \cos(n\omega_0 t) dt$$

$$a_n = \frac{2}{T} \int_0^T \frac{At}{T} \cos(n\omega_0 t) dt$$

$$a_n = \frac{2A}{T^2} \left\{ \frac{t \sin(n\omega_0 t)}{n\omega_0} \right|_0^T - \int_0^T \frac{\sin(n\omega_0 t)}{n\omega_0} dt \right\}$$

$$a_n = \frac{2A}{T^2} \left\{ \frac{T \sin\left(\frac{2n\pi}{T} T\right)}{\frac{n2\pi}{T}} + \frac{\cos(n\omega_0 t)}{\left(\frac{n2\pi}{T}\right)^2} \right|_0^T \right\}$$

Step 5 of 6

$$a_n = \frac{2A}{T^2} \left\{ \frac{T^2 \sin(2n\pi)}{2n\pi} + \frac{\cos\left(\frac{2n\pi}{T} T\right) - 1}{\left(\frac{n2\pi}{T}\right)^2} \right\}$$

We know that $\sin(2n\pi) = 0$

$$a_n = \frac{2A}{T^2} \left\{ \frac{\cos(2n\pi) - 1}{\left(\frac{n2\pi}{T}\right)^2} \right\}$$

Since $\cos(2n\pi) = 1$

$$\therefore a_n = 0$$

Step 6 of 6

$$b_n = \frac{2}{T} \int_0^T x(t) \sin(n\omega_0 t) dt$$

$$b_n = \frac{2}{T} \int_0^T \frac{At}{T} \sin\left(\frac{n2\pi}{T}t\right) dt$$

$$b_n = \frac{2A}{T^2} \int_0^T t \sin\left(\frac{n2\pi}{T}t\right) dt$$

$$b_n = \frac{2A}{T^2} \left\{ \left. \frac{-t \cos\left(\frac{n2\pi}{T}t\right)}{\frac{2n\pi}{T}} \right|_0^T + \int_0^T \frac{\cos\left(\frac{n2\pi}{T}t\right)}{\frac{2n\pi}{T}} dt \right\}$$

$$b_n = \frac{2A}{T^2} \left\{ \left. \frac{-T \cos(2n\pi)}{\frac{2n\pi}{T}} + \frac{\sin\left(\frac{2n\pi}{T}t\right)}{\left(\frac{2n\pi}{T}\right)^2} \right|_0^T \right\}$$

$$b_n = \frac{2A}{T^2} \left\{ \frac{-T^2}{2n\pi} + \frac{\sin(2n\pi) - 0}{\left(\frac{2n\pi}{T}\right)^2} \right\}$$

Since $\sin(2n\pi) = 0$

$$\therefore b_n = \frac{-A}{n\pi}$$

Therefore the Fourier series is

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)$$

$$\boxed{x(t) = \frac{A}{2} - \frac{A}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin\left(\frac{2n\pi}{T}t\right)}$$